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(12) **United States Design Patent** (10) **Patent No.:** **US D1,092,772 S**
Shah (45) **Date of Patent:** **** Sep. 9, 2025**

(54) **PLUNGER CAP FOR SAMPLING APPARATUS**

(71) Applicant: **Quest Diagnostics Investments LLC,**
Secaucus, NJ (US)

(72) Inventor: **Asad Shah,** Sylmar, CA (US)

(73) Assignee: **Quest Diagnostics Investments LLC,**
Secaucus, NJ (US)

(**) Term: **15 Years**

(21) Appl. No.: **29/889,697**

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(51) **LOC (15) Cl.** **24-02**

(52) **U.S. Cl.**
USPC **D24/216**

(58) **Field of Classification Search**
USPC D24/216, 222, 223, 224, 226, 227, 229
CPC A61B 10/0051; A61B 10/0096; A61B
10/0045; A61B 5/154; B01L 3/5082;
B01L 3/50825; A61J 1/1468; G01N
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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

5,744,087 A * 4/1998 Williams B29C 43/00
264/294
D422,357 S * 4/2000 Niedospial, Jr. D24/129
D564,879 S * 3/2008 Baughman D9/454
D738,210 S * 9/2015 Tanner D9/440
D861,735 S * 10/2019 Mish D15/9
D902,429 S * 11/2020 Ganesan D24/224
D908,486 S * 1/2021 Marantis A61J 1/1406
D9/439
D965,529 S * 10/2022 He D13/133
2021/0163174 A1 * 6/2021 Iwata B65D 41/02

FOREIGN PATENT DOCUMENTS

CN 303209129 * 5/2015
JP D1066096 * 4/2000
JP D1454611 * 11/2012
JP D1661077 * 6/2020

OTHER PUBLICATIONS

Silicone bung silicone rubber stopper for glass test tube—ETOL,
posted date unavailable [retrieved online Feb. 20, 2025]. Retrieved
from internet, <https://product.etol-rubber.com/all/silicone-bung-silicone-rubber-stopper-for-glass-test-tube> (Year: 2025).*

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Primary Examiner — Justin M Jonaitis

Assistant Examiner — Samantha K Pollack

(74) *Attorney, Agent, or Firm* — Foley & Lardner LLP

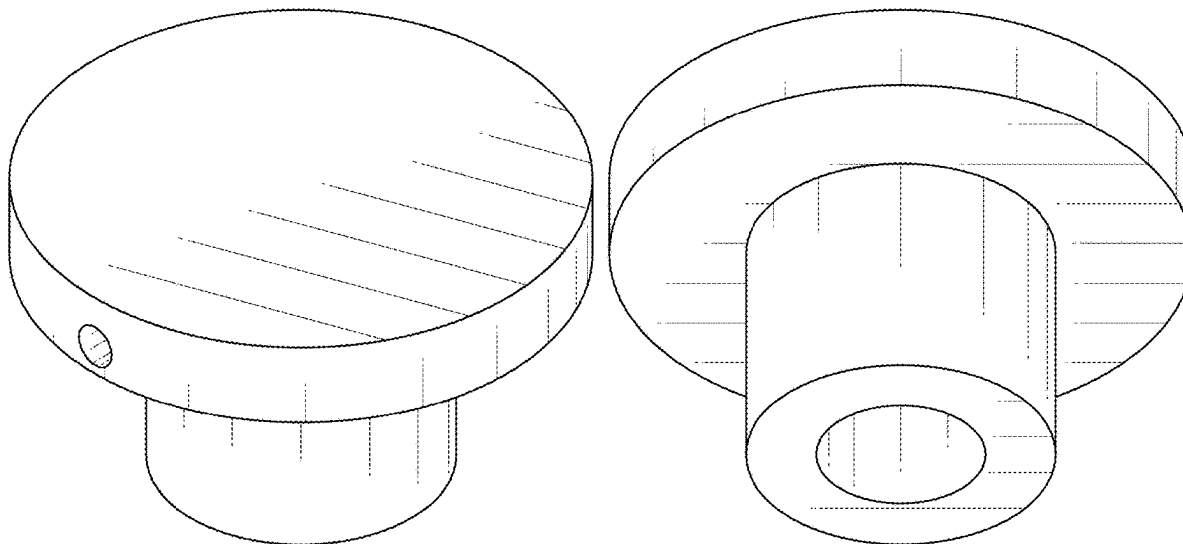
(57) **CLAIM**

The ornamental design for a plunger cap for a sampling
apparatus as shown and described.

DESCRIPTION

FIG. 1 is a front perspective view of the claimed design;
FIG. 2 is a rear perspective view of the claimed design of
FIG. 1;
FIG. 3 is a top view of the claimed design of FIG. 1;
FIG. 4 is a bottom view of the claimed design of FIG. 1;
FIG. 5 is a front view of the claimed design of FIG. 1;
FIG. 6 is a left side view of the claimed design of FIG. 1;
FIG. 7 is a right side view of the claimed design of FIG. 1;
and,
FIG. 8 is a rear view of the claimed design of FIG. 1.
The broken lines in the drawings illustrate portions of the
plunger cap for sampling apparatus that form no part of the
claimed design.

1 Claim, 4 Drawing Sheets



(56)

References Cited

OTHER PUBLICATIONS

General Specimen Collection | Quest Diagnostics, posted date unavailable [retrieved online Feb. 20, 2025]. Retrieved from internet, <https://www.questdiagnostics.com/healthcare-professionals/test-directory/specimen-handling/specimen-collection> (Year: 2025).*

Stopper for 18mm Tube, pk/25 | Hach, posted date unavailable [retrieved online Feb. 20, 2025]. Retrieved from internet, <https://www.hach.com/p-stopper-for-18mm-tube-pk25/173125> (Year: 2025).*

Custom rubber stopper/25mm rubber seal covers/plugs—ETOL, posted date unavailable [retrieved online Feb. 20, 2025]. Retrieved from internet, <https://product.etol-rubber.com/all/custom-rubber-stopper-25mm-rubber-seal-covers-plugs> (Year: 2025).*

Amazon.com: Silicone Rubber Plug End Caps Inserts, posted Aug. 15, 2023 [retrieved online Feb. 20, 2025]. Retrieved from internet, <https://www.amazon.com/Silicone-Rubber-Plug-Inserts-Stopper/dp/B0BZ4ZR3J9?th=1> (Year: 2025).*

* cited by examiner

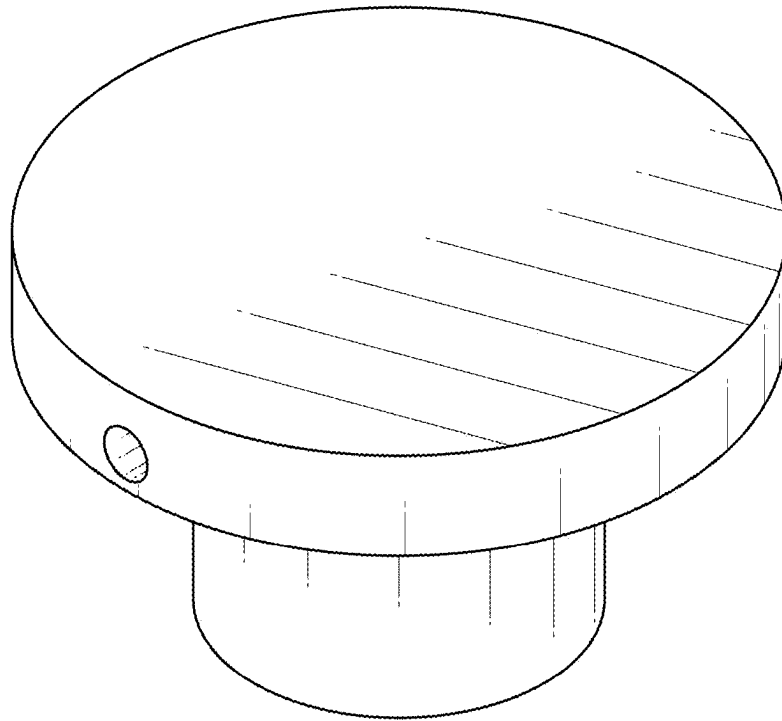


FIG. 1

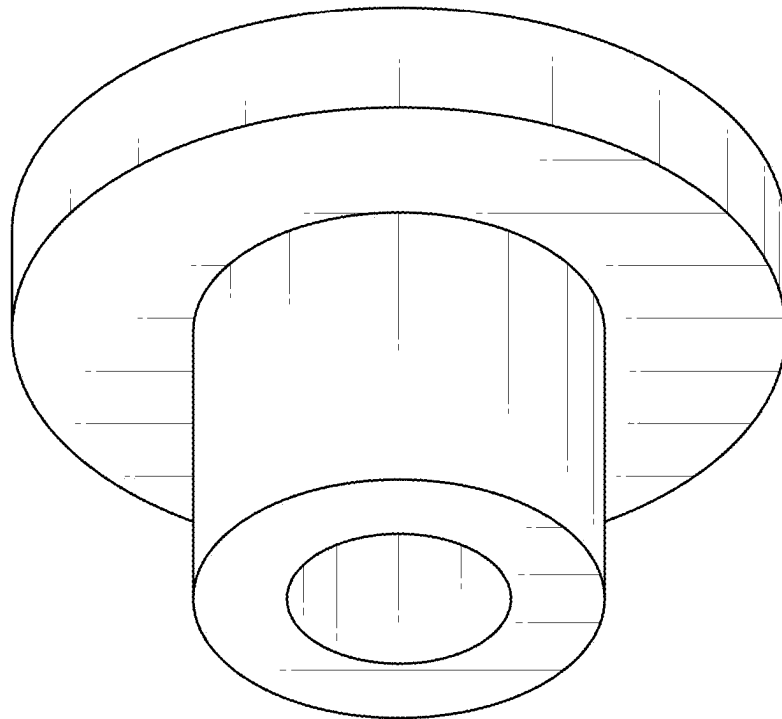


FIG. 2

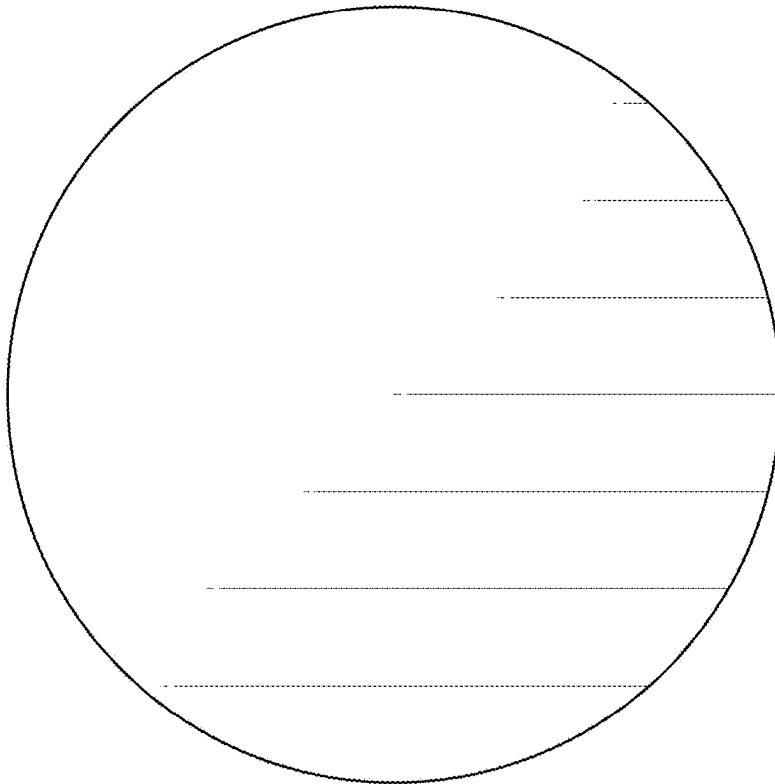


FIG. 3

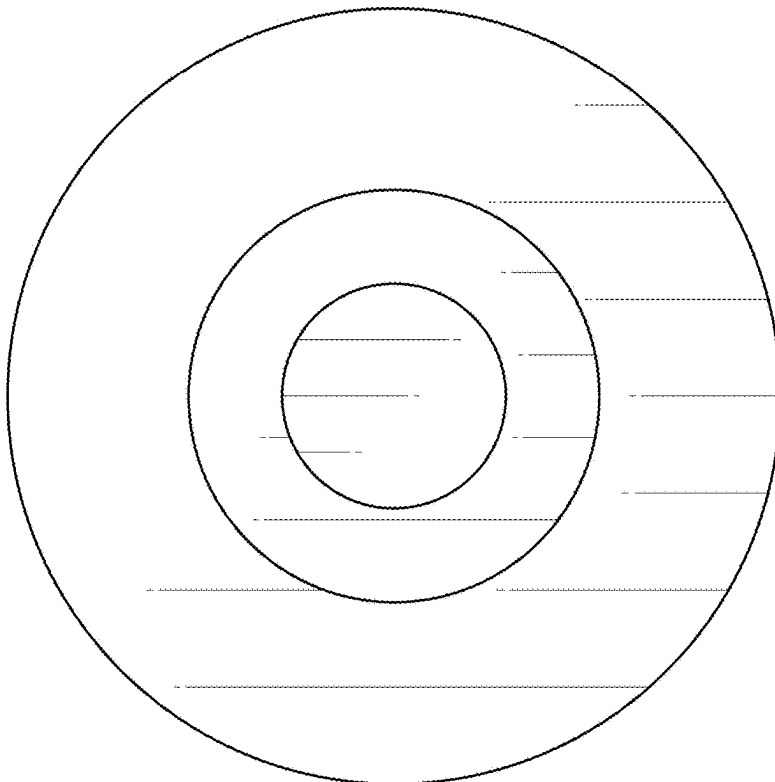


FIG. 4

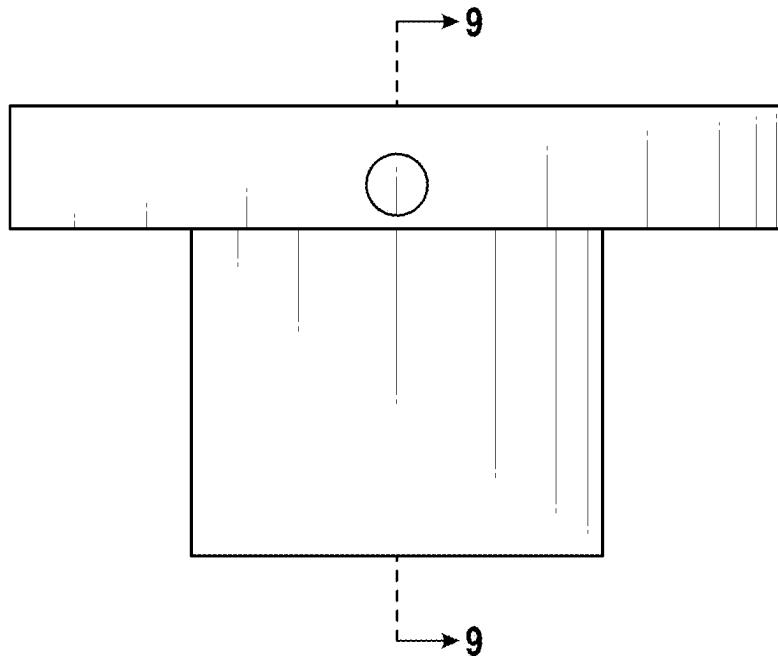


FIG. 5

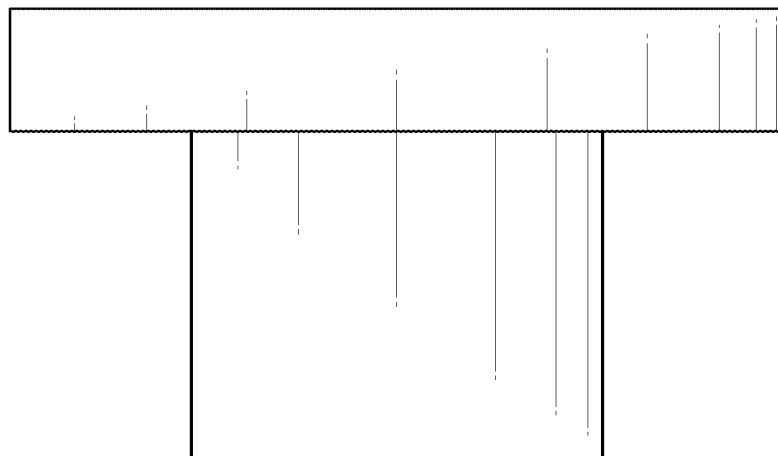


FIG. 6

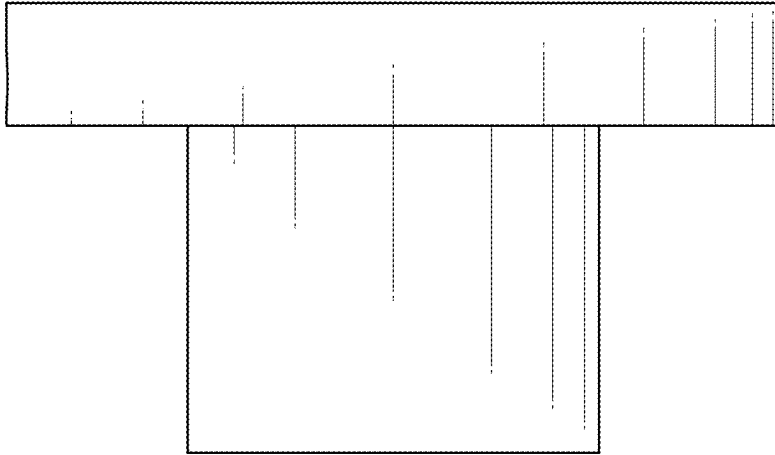


FIG. 7

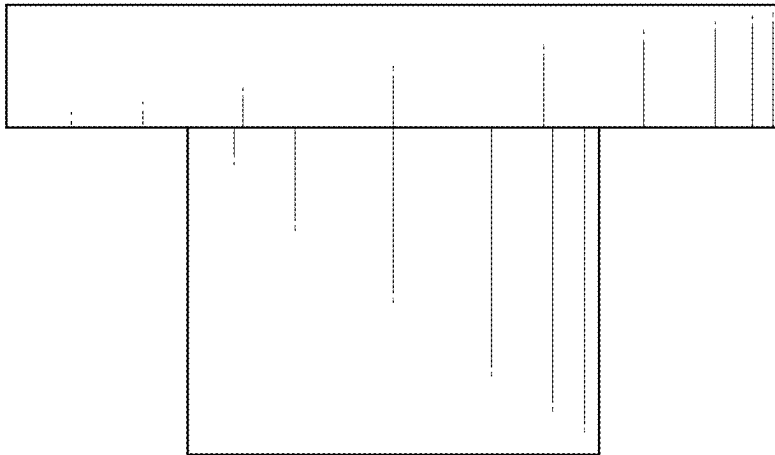


FIG. 8